

FIN 201 - LECTURE 6: TIME VALUE OF MONEY AND LOAN AMORTISATION

These notes accompany the sixth lecture of the introductory corporate finance course. They cover discounting, annuities, the construction of a loan amortisation schedule, and the difference between nominal and effective interest rates. Worked examples use Indian rupee amounts.

1. Why money has a time value

A rupee today is worth more than a rupee next year because it can be invested to earn a return, because prices rise, and because the future is uncertain. The interest rate is the exchange rate between money now and money later. Discounting converts a future amount into its present value; compounding does the reverse.

The present value of a single amount F received after n periods at a periodic rate r is $PV = F / (1 + r)^n$. The factor $1 / (1 + r)^n$ is called the discount factor. As either r or n increases, the discount factor falls.

2. Annuities

An annuity is a level stream of payments at regular intervals. The present value of an ordinary annuity of C per period for n periods is $PV = C \times [1 - (1 + r)^{-n}] / r$. This formula is derived by summing a geometric series of discount factors. An annuity due pays at the start of each period and is worth $(1 + r)$ times an ordinary annuity.

Worked example: a scholarship pays Rs. 10,000 at the end of each year for 5 years and the discount rate is 8%. $PV = 10,000 \times [1 - 1.08^{-5}] / 0.08 = 10,000 \times 3.9927 = \text{Rs. } 39,927$.

3. The EMI formula

A loan repaid in equal monthly instalments is an annuity in reverse: the lender gives the borrower the present value today and receives the level payments. Rearranging the annuity formula gives the instalment: $EMI = P \times r \times (1 + r)^n / [(1 + r)^n - 1]$, where P is the principal, r the monthly rate (annual rate divided by 12) and n the number of months.

Worked example: $P = \text{Rs. } 10,00,000$, annual rate 10%, $n = 60$. Then $r = 0.008333$, $(1 + r)^{60} = 1.6453$, and $EMI = 10,00,000 \times 0.008333 \times 1.6453 / 0.6453 = \text{Rs. } 21,247$. Total repayment is $60 \times 21,247 = \text{Rs. } 12,74,820$, so total interest is about Rs. 2,74,820.

4. Building an amortisation schedule

Each instalment is split into interest and principal. Interest for the month equals the opening balance times r ; the principal component is the EMI minus that interest; the closing balance is the opening balance minus the principal component. Early instalments are mostly interest, later ones mostly principal. A spreadsheet with five columns - month, opening balance, interest, principal, closing balance - reproduces the whole schedule, and the closing balance after the final month should be zero.

Common student error: computing interest on the original principal every month. That is the flat-rate method used by some lenders, and it overstates interest by roughly 45 to 50 percent for a five-year loan compared with the reducing-balance method.

5. Nominal versus effective rates

A nominal annual rate of 12% compounded monthly is really 1% per month, and $(1.01)^{12} - 1 = 12.68\%$ per year. The effective annual rate (EAR) is what the borrower actually pays when compounding is taken

into account. When a loan carries upfront fees, the borrower receives less than the principal but repays the full schedule; the internal rate of return of those cash flows is the true cost, and it is always above the nominal rate.

Worked example: a fee of 2% on a one-year loan at 12% nominal raises the effective cost to roughly 14.3%, because the borrower receives Rs. 98 for every Rs. 100 repaid with interest.

6. Summary and reading

Discounting and compounding are inverses. Annuity formulas price level cash-flow streams, and the EMI formula is the annuity formula solved for the payment. Amortisation schedules show how each instalment splits. Effective rates, not nominal rates, are the basis for comparing offers. Read Brealey, Myers and Allen chapter 2 and attempt problems 6.1 to 6.8 before the tutorial.

Practice questions

Q1. A loan of Rs. 5,00,000 at 9% per annum for 36 months: compute the EMI and the interest in the first month. Q2. Explain why an annuity due is worth more than an ordinary annuity. Q3. A lender advertises 9.5% flat on a 60-month vehicle loan; estimate the reducing-balance equivalent. Q4. Why does a processing fee raise the effective rate more on a short loan than on a long one?